

BANK LOAN REPORT SQL QUERY ANALYSIS

A. BANK LOAN REPORT | SUMMARY

KPI's:

1. Total Loan Applications

```
SELECT COUNT(id) AS Total_Loan_Applications FROM bank_loan_data
```

	MTD_Total_Amount_Received
1	58074380

Insight: The bank processed over 38,000 loan applications, indicating strong lending activity and a large customer base.

2. MTD Loan Applications

```
SELECT COUNT(id) AS Total_Loan_Applications FROM bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Loan_Applications
1	4314

Insight: December recorded 4,314 applications, showing sustained customer demand for loans at year-end.

3. PMTD Loan Applications

```
SELECT COUNT(id) AS PMTD_Loan_Applications FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Loan_Applications
1	4035

Insight: Applications increased from 4,035 in November to 4,314 in December, reflecting month-over-month growth in lending demand.

4. Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount FROM bank_loan_data
```

	Total_Funded_Amount
1	435757075

Insight: The bank disbursed over \$435 million in loans, demonstrating a significant lending portfolio.

5. MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Total_Funded_Amount
1	53981425

Insight: December funding exceeded \$53.9 million, representing the highest monthly loan disbursement of the year.

6. PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Total_Funded_Amount
1	47754825

Insight: Loan funding increased from November to December, indicating expansion in lending operations.

7. Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Received FROM bank_loan_data
```

	Total_Amount_Received
1	473070933

Insight: The bank recovered over \$473 million from borrowers, exceeding the total funded amount and generating positive returns.

8. MTD Total Amount Received

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Received FROM bank_loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Total_Amount_Received
1	58074380

Insight: Loan repayments reached \$58 million in December, highlighting strong collection performance.

9. PMTD Total Amount Received

```
SELECT SUM(total_payment) AS PMTD_Total_Amount_Received FROM bank_loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Total_Amount_Received
1	50132030

Insight: Repayment collections increased month-over-month, suggesting improving cash inflows.

10. Average Interest Rate

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS Average_Interest_Rate FROM bank_loan_data
```

	Average_Interest_Rate
1	12.05

Insight: The portfolio maintained an average interest rate of approximately 12%, balancing profitability and borrower affordability.

11. MTD Average Interest

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS MTD_Average_Interest_Rate FROM bank_loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Average_Interest_Rate
1	12.36

Insight: December's average interest rate rose slightly, indicating a shift toward higher-risk or higher-yield loans.

12. PMTD Average Interest

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS PMTD_Average_Interest_Rate FROM bank_loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Average_Interest_Rate
1	11.94

Insight: Interest rates remained relatively stable, reflecting consistent lending policies.

13. Average DTI

```
SELECT ROUND(AVG(dti), 4) * 100 AS Average_DTI FROM bank_loan_data
```

	Average_DTI
1	13.33

Insight: The average borrower DTI of 13.33% suggests borrowers generally maintained manageable debt levels.

14. MTD Average DTI

```
Average ROUND(AVG(dti), 4) * 100 AS MTD_Average_DTI FROM bank_loan_data
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Average_DTI
1	13.67

Insight: December borrowers exhibited slightly higher leverage, requiring continued credit risk monitoring.

15. PMTD Average DTI

```
SELECT ROUND(AVG(dti), 4) * 100 AS PMTD_Average_DTI FROM bank_loan_data
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Average_DTI
1	13.3

Insight: DTI remained stable across months, indicating consistent borrower financial profiles.

GOOD LOAN ISSUED

1. Good Loan Percentage

```
SELECT
  (COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id
END) * 100) /
  COUNT(id) AS Good_Loan_Percentage
FROM bank_loan_data
```

	Good_Loan_Percentage
1	86

Insight: Approximately 86% of all loans were classified as Good Loans, demonstrating a healthy and well-performing loan portfolio.

2. Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Applications
1	33243

Insight: More than 33,000 applications fell into the Good Loan category, indicating strong borrower repayment behavior.

3. Good Loan Funded Amount

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Funded_Amount
1	370224850

Insight: The majority of the bank's lending capital was allocated to low-risk borrowers.

4. Good Loan Amount Received

```
SELECT SUM(total_payment) AS Good_Loan_Received_Amount FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Received_Amount
1	435786170

Insight: Good Loans generated the largest share of repayments, serving as the primary revenue driver of the portfolio.

BAD LOAN ISSUED

1. Bad Loan Percentage

```
SELECT
  (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0) /
  COUNT(id) AS Bad_Loan_Percentage
FROM bank_loan_data
```

	Bad_Loan_Percentage
1	13.824657818332

Insight: Only 13.8% of loans were charged off, suggesting overall credit risk remains under control.

2. Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_Applications
1	5333

Insight: Around 5,300 loans defaulted, highlighting areas where underwriting policies could be strengthened.

3. Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_Amount FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_Funded_Amount
1	65532225

Insight: Over \$65 million was funded to borrowers who eventually defaulted, representing potential credit losses.

4. Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_Received_Amount FROM bank_loan_data
WHERE loan_status = 'Charged Off'
```

	Bad_Loan_Received_Amount
1	37284763

Insight: Recoveries from charged-off loans were significantly lower than funded amounts, confirming their negative impact on profitability.

LOAN STATUS

1. Loan Status Summary

```
SELECT
    loan_status,
    COUNT(id) AS Total_Loan_Applications,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Interest_Rate,
    Avg(dti * 100) AS DTI
FROM bank_loan_data
GROUP BY loan_status
```

	loan_status	Total_Loan_Applications	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

Insight: Fully Paid loans accounted for the largest share of applications, funded amounts, and repayments. Charged-Off loans carried the highest risk profile with higher interest rates and DTI ratios.

2. MTD Loan Status Analysis

```
SELECT
    loan_status,
    SUM(total_payment) AS MTD_Total_Amount_Received,
    SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
GROUP BY loan_status
```

	loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
1	Fully Paid	47815851	41302025
2	Charged Off	5324211	8732775
3	Current	4934318	3946625

Insight: Fully Paid loans continued to dominate December's portfolio, while Charged-Off loans represented a relatively small portion of monthly activity.

BANK LOAN REPORT | OVERVIEW

1. Monthly Trend Analysis

```
SELECT
    MONTH(issue_date) AS Month_Number,
    DATENAME(MONTH, issue_date) AS Month_Name,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY MONTH(issue_date)
```

	Month_Number	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

Insight: Loan applications, funding amounts, and repayments increased steadily throughout the year, with December recording the strongest performance across all three metrics.

2. State Analysis

```
SELECT
    address_state,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY address_state
ORDER BY address_state
```

	address_state	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660
24	MN	592	6302600	6750746
25	MO	660	7151175	7692732
26	MS	19	139125	149342
27	MT	79	829525	892047
28	NC	759	8787575	9534813
29	NE	5	31700	24542
30	NH	161	1917900	2101386
31	NJ	1822	21657475	23425159
32	NM	183	1916775	2084485
33	NV	482	5307375	5451443
34	NY	3701	42077050	46108181
35	OH	1188	12991375	14330148
36	OK	293	3365725	3712649
37	OR	436	4720150	4966903
38	PA	1482	15826525	17462908
39	RI	196	1883025	2001774
40	SC	464	5080475	5462458
41	SD	63	606150	656514
42	TN	17	162175	141522
43	TX	2664	31236650	34392715
44	UT	252	2849225	2952412
45	VA	1375	15982650	17711443
46	VT	54	504100	534973
47	WA	805	8855525	9531739
48	WI	446	5070450	5485161
49	WV	167	1830525	1991936
50	WY	79	890750	1046050

Insight: California, New York, Texas, and Florida generated the highest lending activity, making them the bank's most important regional markets.

3. Loan Term Analysis

```
SELECT
    term AS Term,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY term
ORDER BY term
```

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

Insight: 36-month loans represented nearly three times more applications than 60-month loans, indicating borrower preference for shorter repayment periods.

4. Employee Length Analysis

```
SELECT
    emp_length AS Employee_Length,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length
```

	Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

Insight: Borrowers with 10+ years of employment received the highest loan funding and exhibited the strongest repayment contribution, suggesting employment stability is a key indicator of creditworthiness.

5. Loan Purpose Analysis

```
SELECT
    purpose AS Purpose,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose
```

	Purpose	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	car	1497	10223575	11324914
2	credit card	4998	58885175	65214084
3	Debt consolidation	18214	232459675	253801871
4	educational	315	2161650	2248380
5	home improvement	2876	33350775	36380930
6	house	366	4824925	5185538
7	major purchase	2110	17251600	18676927
8	medical	667	5533225	5851372
9	moving	559	3748125	3999899
10	other	3824	31155750	33289676
11	renewable_energy	94	845750	898931
12	small business	1776	24123100	23814817
13	vacation	352	1967950	2116738
14	wedding	928	9225800	10266856

Insight: Debt Consolidation was the most common borrowing purpose, accounting for the largest share of applications, funding, and repayments. This indicates customers primarily use loans to manage existing debt obligations.

6. Home Ownership Analysis

```
SELECT
    home_ownership AS Home_Ownership,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Received_Amount
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY home_ownership
```

	Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Received_Amount
1	MORTGAGE	17198	219329150	238474438
2	NONE	3	16800	19053
3	OTHER	98	1044975	1025257
4	OWN	2838	29597675	31729129
5	RENT	18439	185768475	201823056

Insight: Customers with mortgages represented the largest borrower segment, followed by renters. Mortgage holders generated the highest funded and recovered amounts, making them the most valuable customer group within the portfolio.

OVERALL BUSINESS CONCLUSION

The analysis reveals a healthy loan portfolio with strong growth in applications, funding, and repayments throughout the year. With 86% of loans classified as good loans, the bank demonstrates effective risk management. Debt consolidation emerged as the primary borrowing need, while mortgage holders and long-tenured employees represented the most reliable customer segments. Regional lending activity was concentrated in major states such as California, New York, Texas, and Florida, offering key opportunities for future business expansion.